

Objective

Create a ROS2 package or a script through which you can give a goal position and orientation for the end effector of the UR5e robot arm, and make the UR5e robot arm reach that position and orientation.

Background

1. In Mowito, you would need to work on the UR5e robot arm. To make the UR5e robot arm do your task at the fundamental level, you will have to make its end-effector reach a certain position and orientation.
2. In order to develop the scripts to control the robot arm, you do not need an actual physical robot arm to be with you. You can use ROS-based simulators and visualizers.

Task

1. Download the correct ROS2 drivers for UR5e.
2. Create a Python script that takes input from the user through the command line for goal position and orientation of the end-effector.
3. The script should be able to talk to the ROS2 drivers of UR5 and make it give commands to the UR5e robot to reach goal position and orientation.
4. Since you do not have the actual hardware, visualize the motion of the robot in the correct visualizer.

Stage 1 - Mandatory

1. Visualize the motion in Rviz

Stage 2 - Bonus

1. Simulate and visualize the motion in Isaac Sim.

Deliverables

1. Link to the Github Repo
 - a. Push the code in a private github repo
 - b. Give the access of that repo to puru07
 - c. Share the link of the github repo over email
2. Google drive LINK (public) to the Screencast videos. These videos should include
 - a. You launching the command

- b. User (you) giving the goal coordinates on the terminal
- c. The visualization of the robot going from initial to goal position and orientation.

Judging Criteria

We will be talking about the script in the interview, and judge you on your ability to

1. explain what's happening in your code
2. When will it break and what are its limitation
3. What will happen when you put it on the real robot.